particles, or four, or whatever. Just as we observe Schrödinger's cat to be either awake or asleep, and not any superposition thereof, collections of particles are what we observe when we make measurements of gently vibrating quantum fields. In the language of the previous chapter, as long as the fields aren't fluctuating too wildly, the "pointer states" of quantum field theory look like collections of definite numbers of particles. Those are the kinds of states we see when we actually look at the world.

Even better, quantum field theory can describe transitions between states with different numbers of particles, just as an electron can hop up or down in energy in an atom. In ordinary particle-based quantum mechanics, the number of particles is fixed, but quantum field theory has no problem describing particles decaying or annihilating or being created in collisions. Which is good, because things like that happen all the time.

Quantum field theory represents one of the great triumphs of unification in the history of physics, tying together the seemingly opposed ideas of particles and waves. Once we realize that quantizing the electromagnetic field leads to particle-like photons, perhaps it shouldn't be surprising that other particles such as electrons and quarks also arise from quantized fields. Electrons are vibrations in the electron field, various types of quarks are vibrations in various types of quark fields, and so on.

Introductions to quantum mechanics sometimes contrast particles and waves as if they are two equal sides of the same coin, but ultimately the battle between particles and fields is not a fair fight. Fields are more fundamental; it's fields that provide the best picture we currently have of what the universe is made of. Particles are simply what we see when we observe fields under the right circumstances. Sometimes the circumstances aren't right; inside a proton or neutron, even though we often speak about quarks and gluons as if they're individual particles, it's more accurate to think of them as diffuse fields. As physicist Paul Davies once titled a paper, with only a bit of rhetorical exaggeration, "Particles Do Not Exist."

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Our interest here is in the basic paradigm of quantum reality, not in the specific pattern of particles and their masses and interactions. We care about entanglement and emergence and how the classical world arises from the